

Tests and Homeworks

Balázs Lengyel

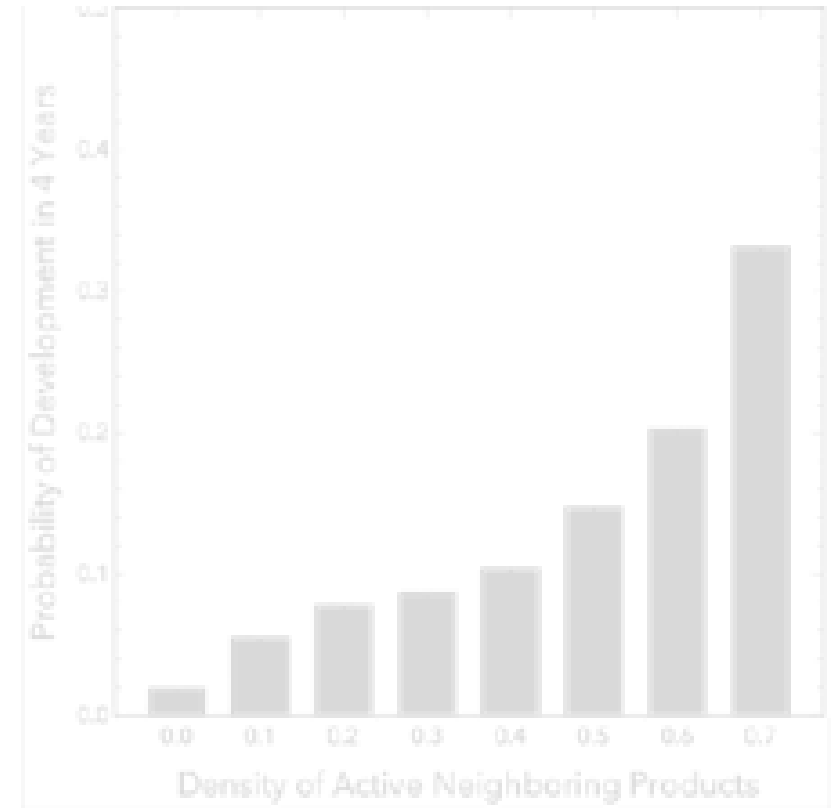
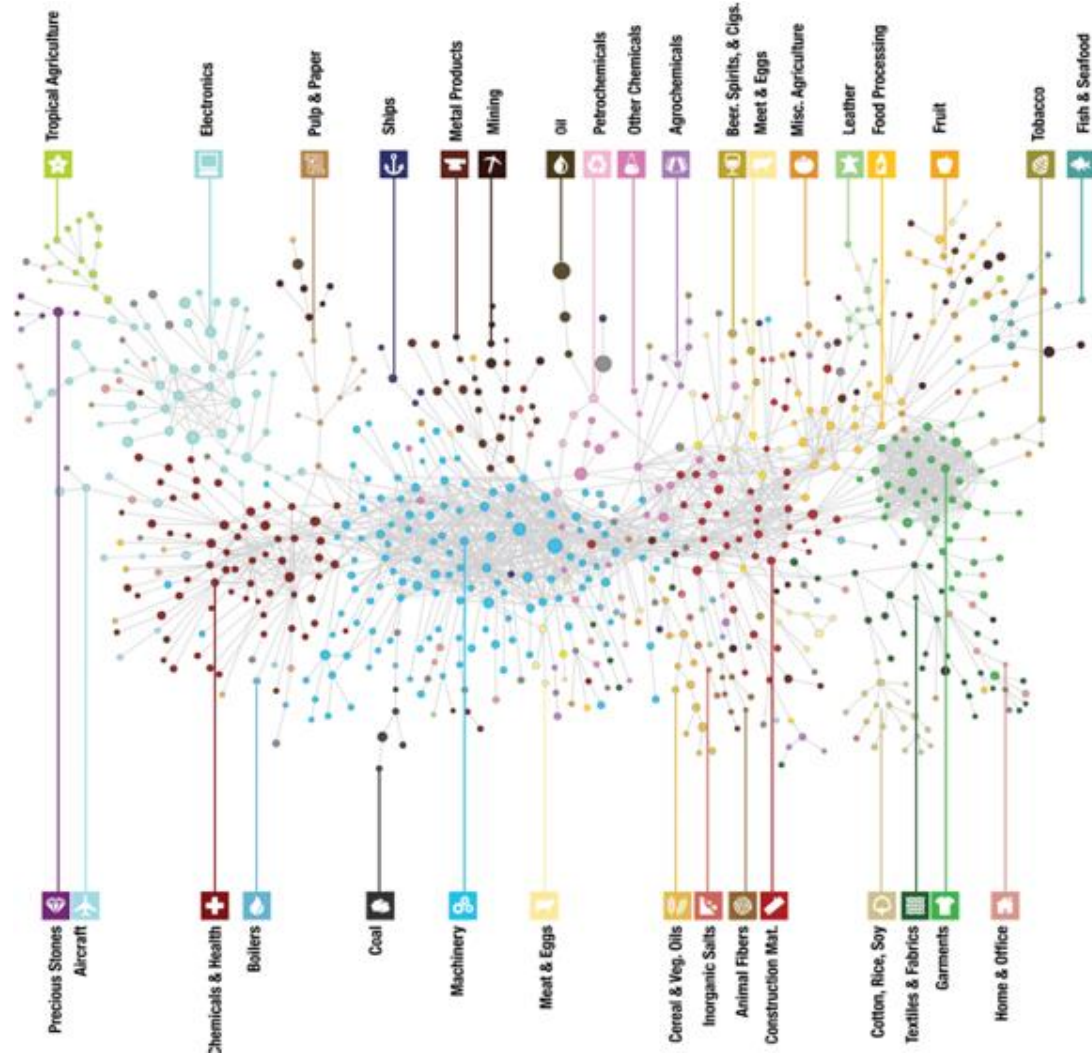
Economic Complexity, Class 2 – Sept 23 2026

Social Data Science, Corvinus University of Budapest

Quiz 1 – Sept 23

1. What are the assumptions behind the product space? Describe at least 2. (2 points)
2. What are the empirical observations made with the use of product space regarding diversification? (1 point)

Observations



Assumptions

Statements that are not verified but the measurement and conclusion rely on them

- RCA can measure capabilities
- Products frequently exported together with $RCA > 1$ share similar capabilities

HW1 for Sept 28

Interesting problem and data.

1. Identify a topic that is connected to the course and interests you. This can be important for a company, city, an industry, a country, for specific groups of people or the global humanity (eg. automation, inequality, international competition, climate change, populism etc.).
2. Search for journal articles that address this topic with interesting data. Choose one that you would like to show to the class.
3. Prepare 1 pptx slide that summarizes the paper. The presentation cannot be longer than 1 slide. Best slides typically include figures that can illustrate the story.
4. Be able to present the slide and talk about the research problem, question, the data, methods, and findings. Save the file: "givenname_familyname_EC2026_HW1.pptx".
5. Send it to the instructor and TA by email. Deadline: September 29, Tuesday, 12noon. Homework sent after the deadline are not considered.
6. Homeworks covering 1-2-3 can get 2 points. Presentation gives an extra 1 point.

Reading for – Sept 30

Hausmann and Hidalgo 2009 PNAS.

Extra homework for Sept 30

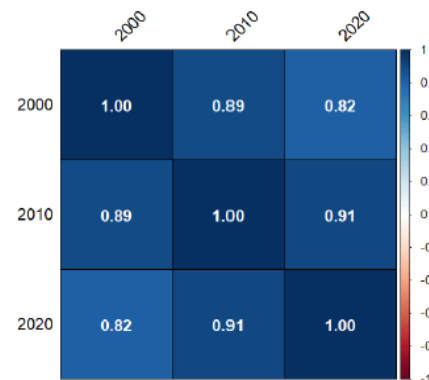
Illustrate the dynamics of complexity ranking in trade, technology and research.

1. Go to the country Complexity Rankings on oec.world and download the country data.
2. Investigate ranking dynamics in all dimensions across 2000-2010-2020. Compare correlations across time but within dimensions. Check cross-correlations across dimensions.
3. /Plus: Merge data with Economic Growth forecast tables <https://oec.world/en/resources/methods/eci-growth-predictions>. Calculate the coefficient of ECI indices using the Solow model of economic growth.
4. Prepare visualizations and discuss the results. Collect figures and text to a pdf (max 2 pages) and save it "givenname_familyname_EC2026_HW1_extra.pdf".
5. Include code and prompts in a zip file "givenname_familyname_EC2026_HW1_documentation.zip".
6. Send it to the instructor and TA by email. Deadline: September 29, Tuesday, 12noon. Homework sent after the deadline are not considered.
7. Extra homeworks covering points 1-2-4-5 can get 2 points. Point 3 gets plus 1 point.

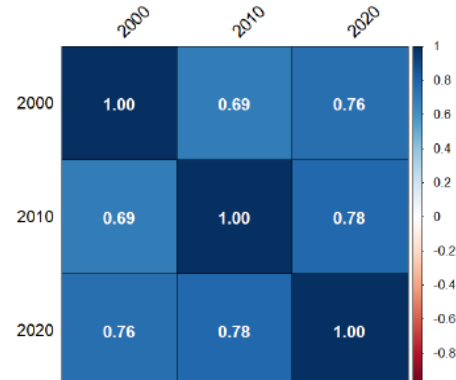
Coloring and caption



1. Figure: Spearman rank correlation matrix for Trade complexity across decades



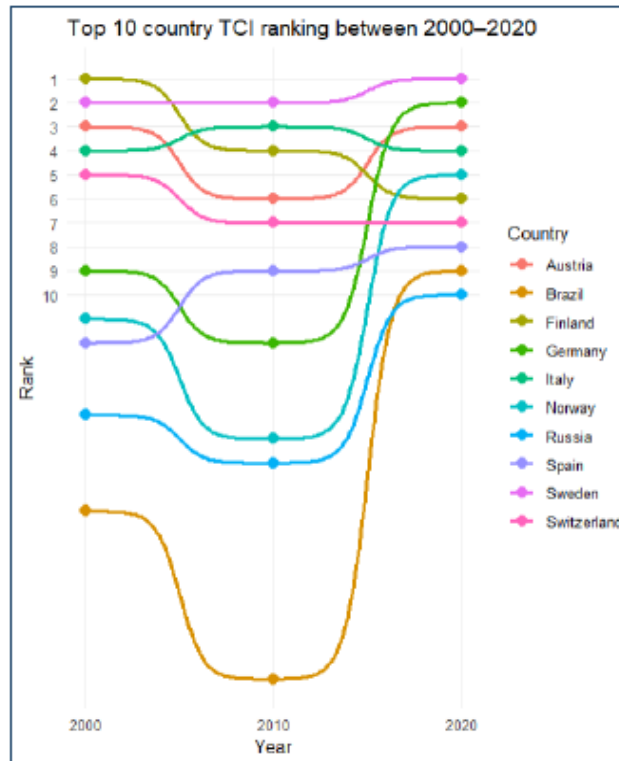
2. Figure: Spearman rank correlation matrix for Technology complexity across decades



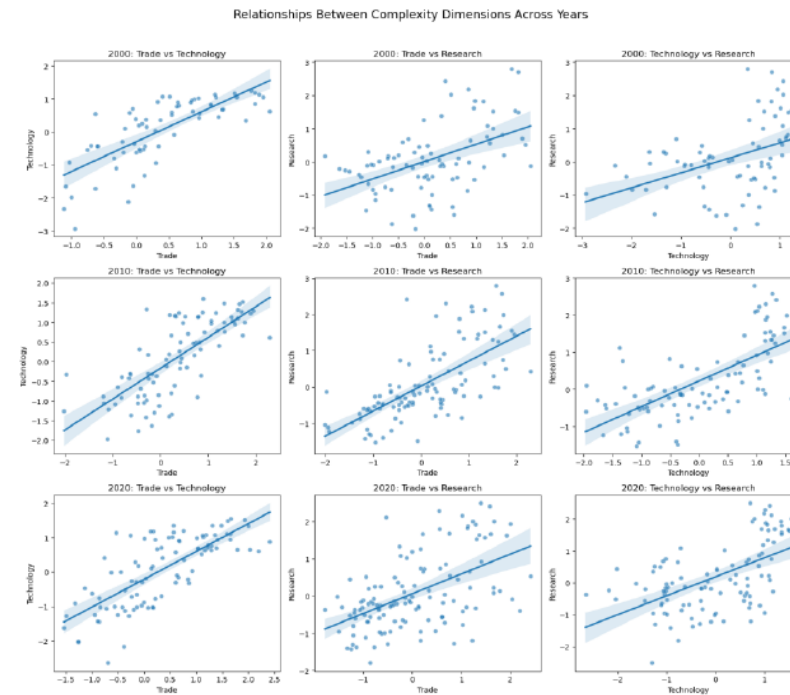
3. Figure: Spearman rank correlation matrix for Research complexity across decades

Period	2000-2010	2010-2020	2000-2020
Trade	0.936	0.940	0.839
Technology	0.900	0.907	0.826
Research	0.706	0.787	0.756

Font size



6. ábra: TCI ranking between 2000-2020



3. Figure Relationships Between Complexity Dimensions Across Years

Regression tables

- Necessary elements: coefficients, significance, constant, N, R2
- Correlation, multicollinearity: VIF tests, separate introduction of variables, interaction terms

	Model
(Intercept)	16.664*** (0.729)
ln(GDP per capita)	-1.506*** (0.073)
ECI Trade	0.587*** (0.074)
ECI Tech	0.665*** (0.075)
ECI Research	-0.089+ (0.048)
Num. Obs.	86
R ²	0.854
R ² Adj.	0.847
p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	

Homework: correlations

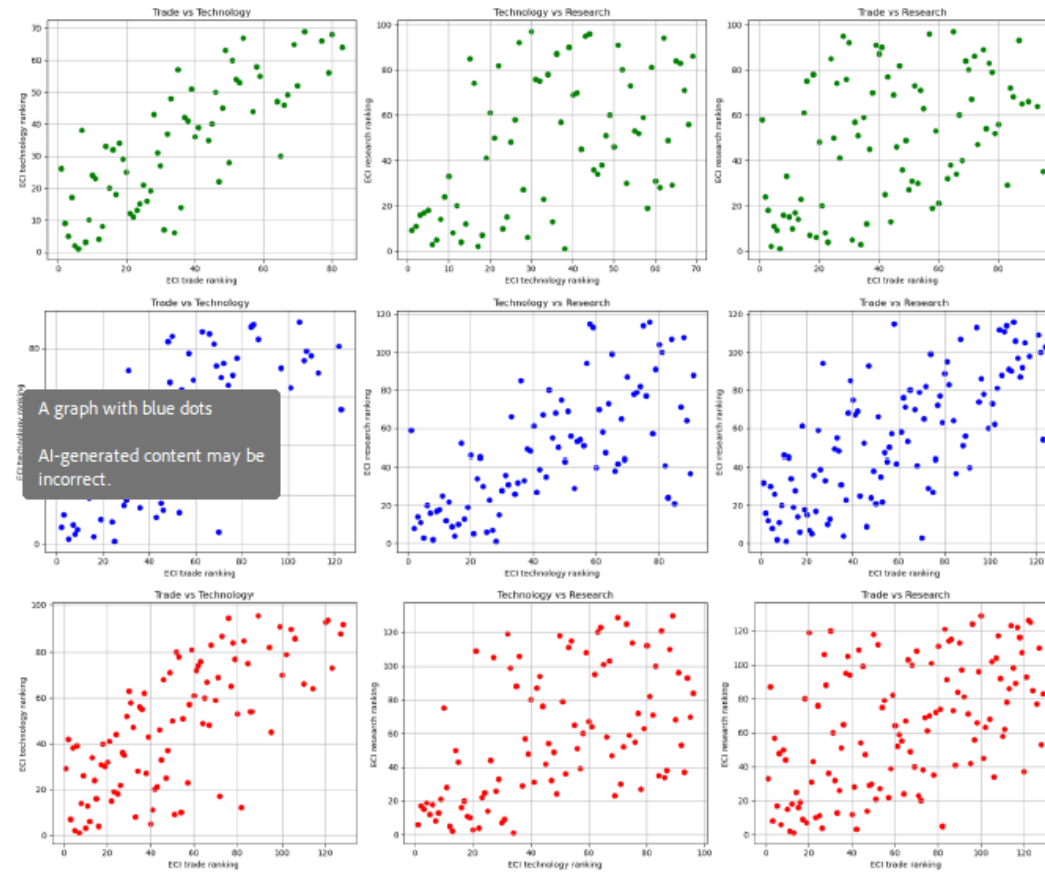


Figure 2: Figure 1: Comparison of ECI rankings between dimensions in 2000, 2010 and 2020

Homework: extras

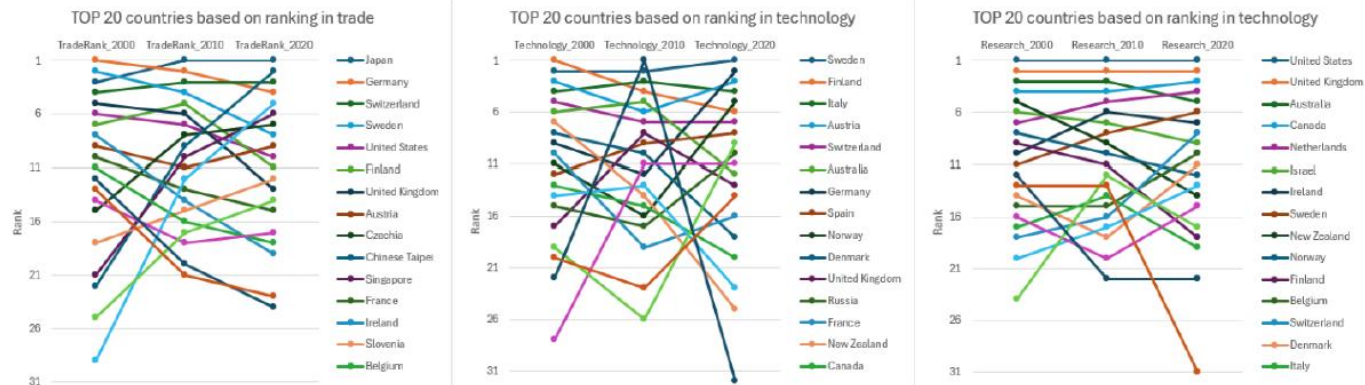


Figure 1: Top 20 countries based on the ranking of country complexity measure in 2000-2010-2020

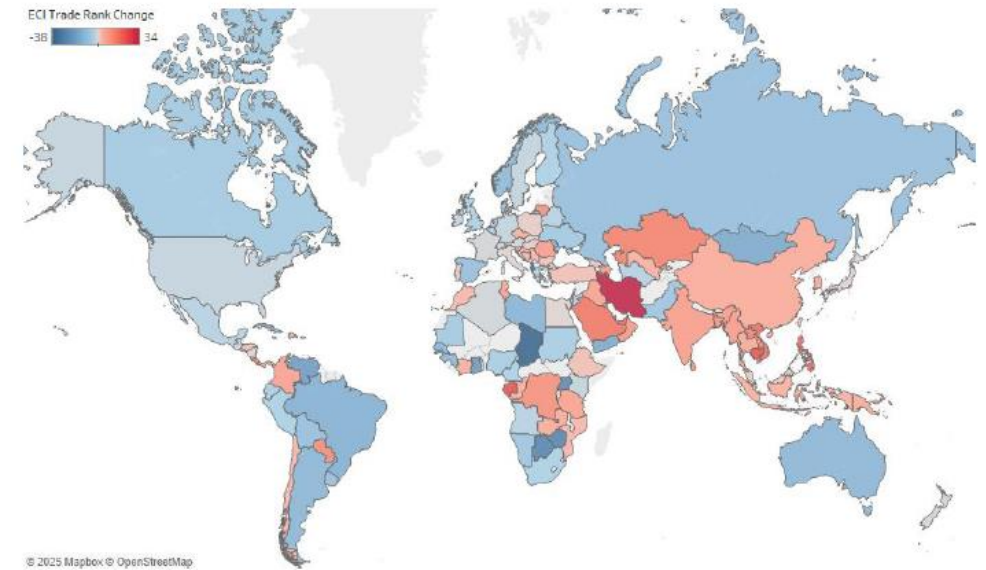
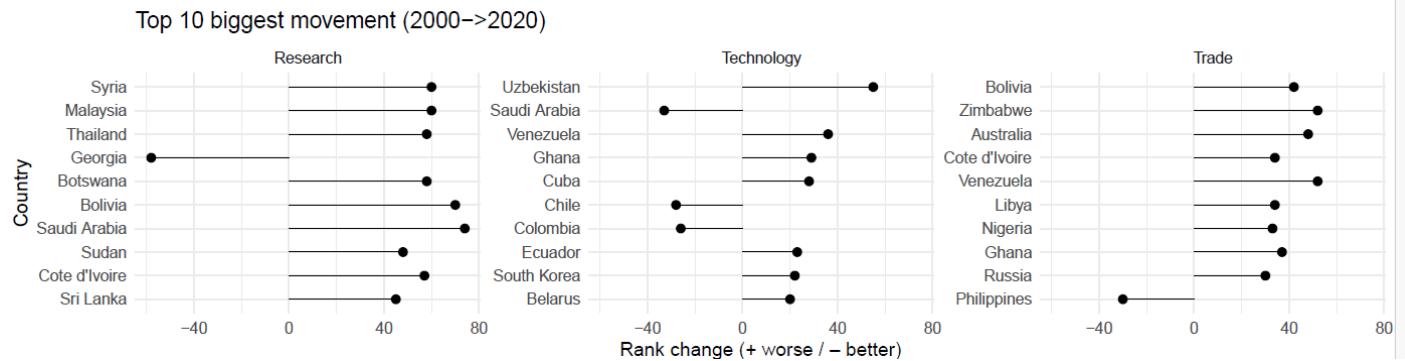
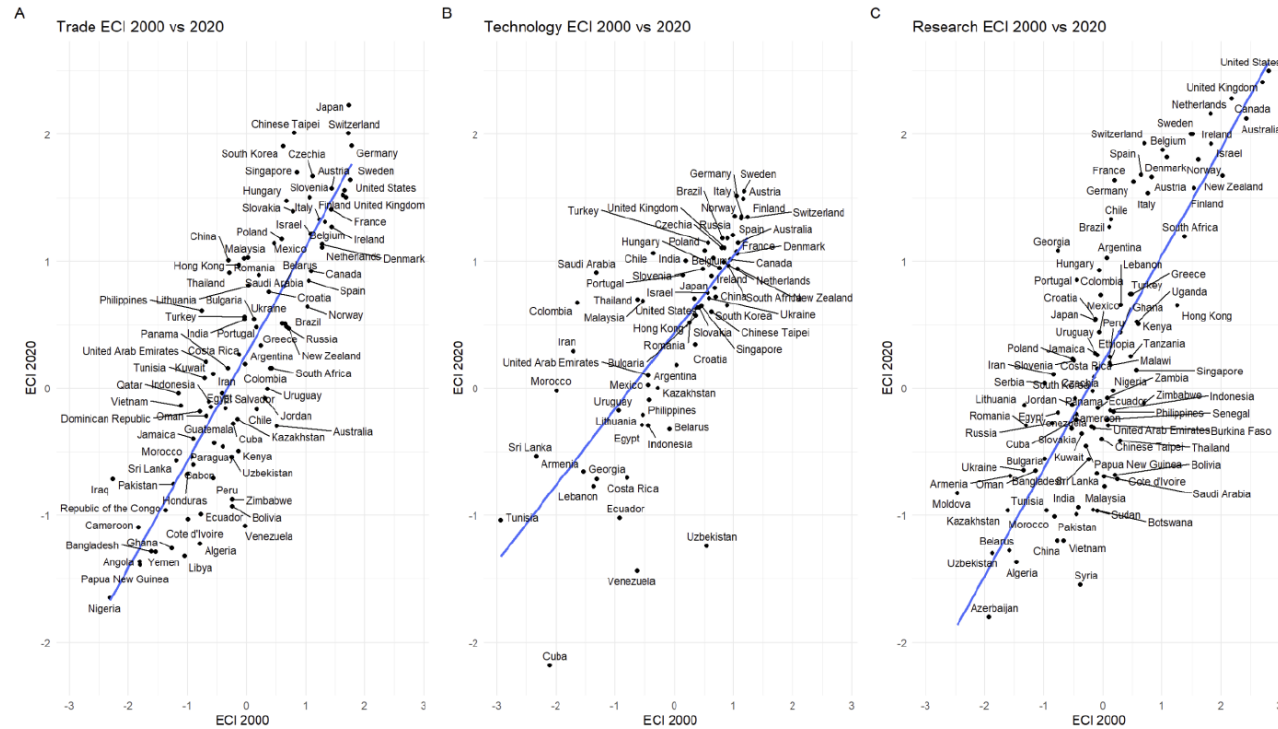
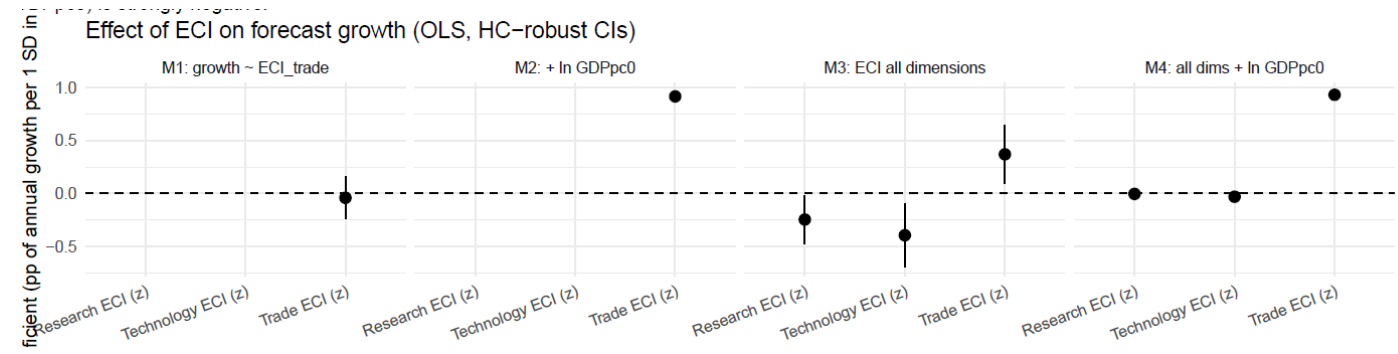
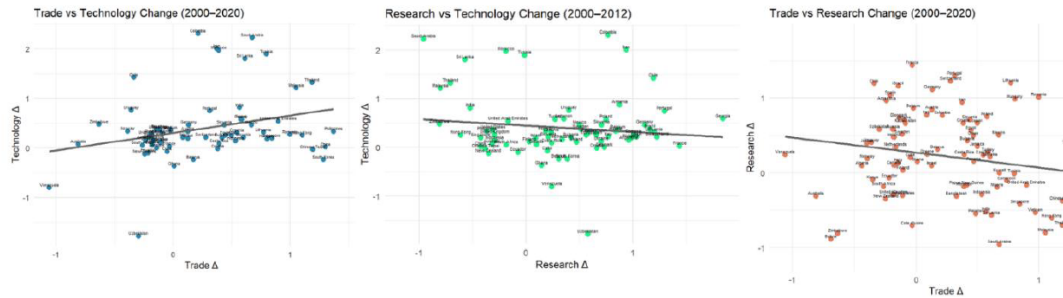


Figure 3: Changes in ECI Trade ranking from 2010 to 2020

Homework: extras

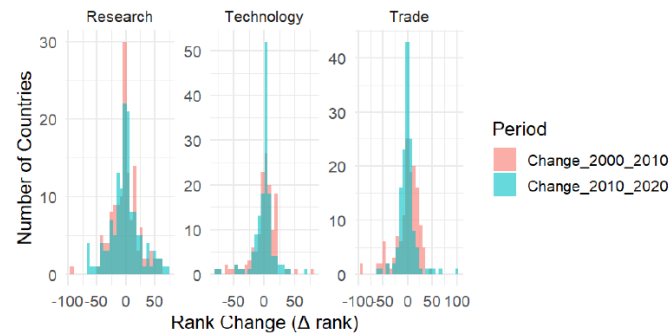


Homework: extras

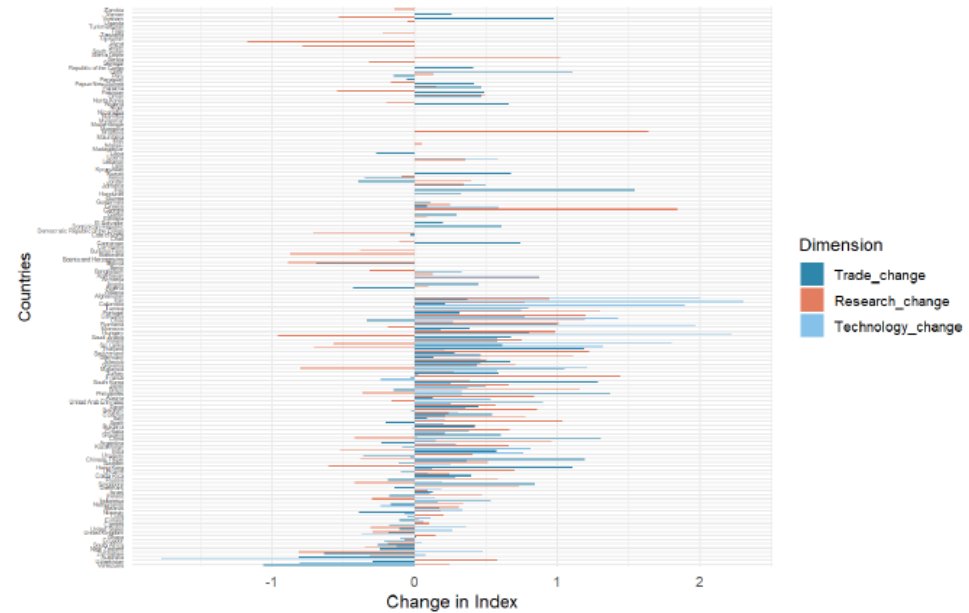


The three scatter plots illustrate how trade, technology, and research complexity changed over time. Between 2000 and 2020, there is a mild positive relationship between trade and technology growth, meaning that countries with more complex trade structures often also advanced technologically. However, the links between research and the other two dimensions are weakly negative, indicating that increases in research complexity did not consistently correspond with progress in trade or technology.

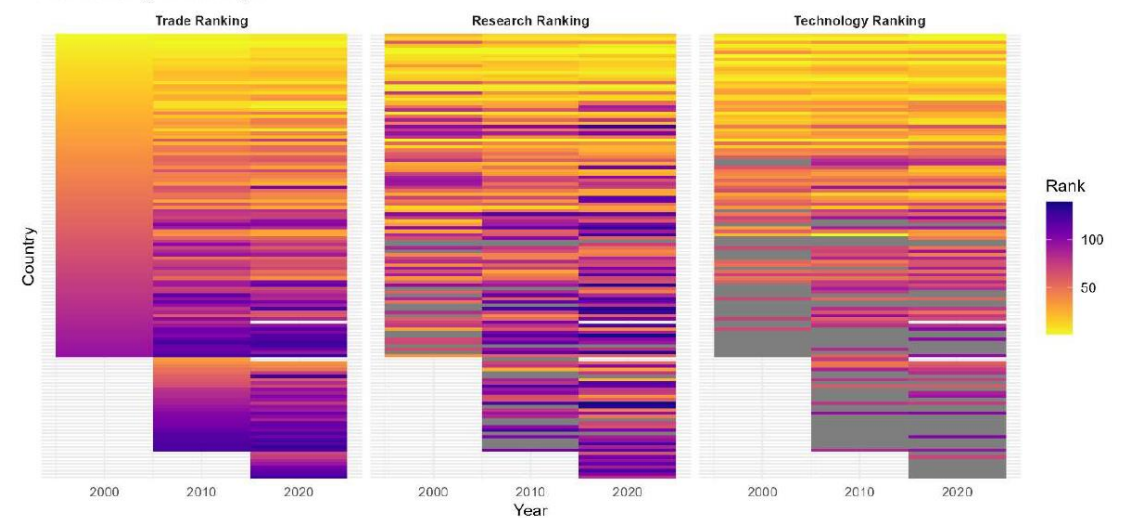
Distribution of Rank Changes (2000–2010–2020)



Change in Economic Complexity (2000–2020)



ECI Rankings Heatmaps



Homework: estimation

Regression Analysis

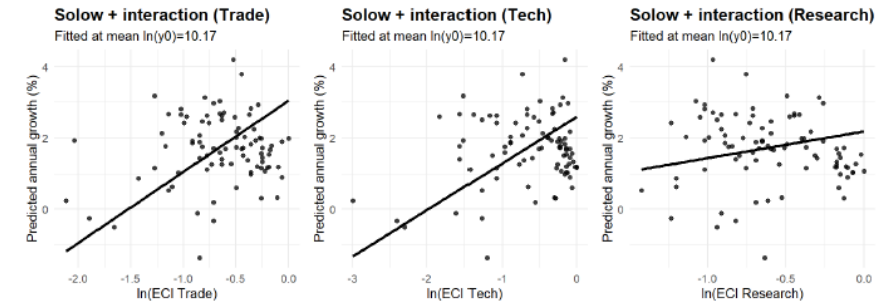


Figure 2

Using the Solow-style specification in Eq. (1), with predicted GDP per capita growth as the outcome and including $\ln(y_0)$, ECI, and their interaction, we find a consistent pattern across dimensions. The patterns can be seen in Figure 2.

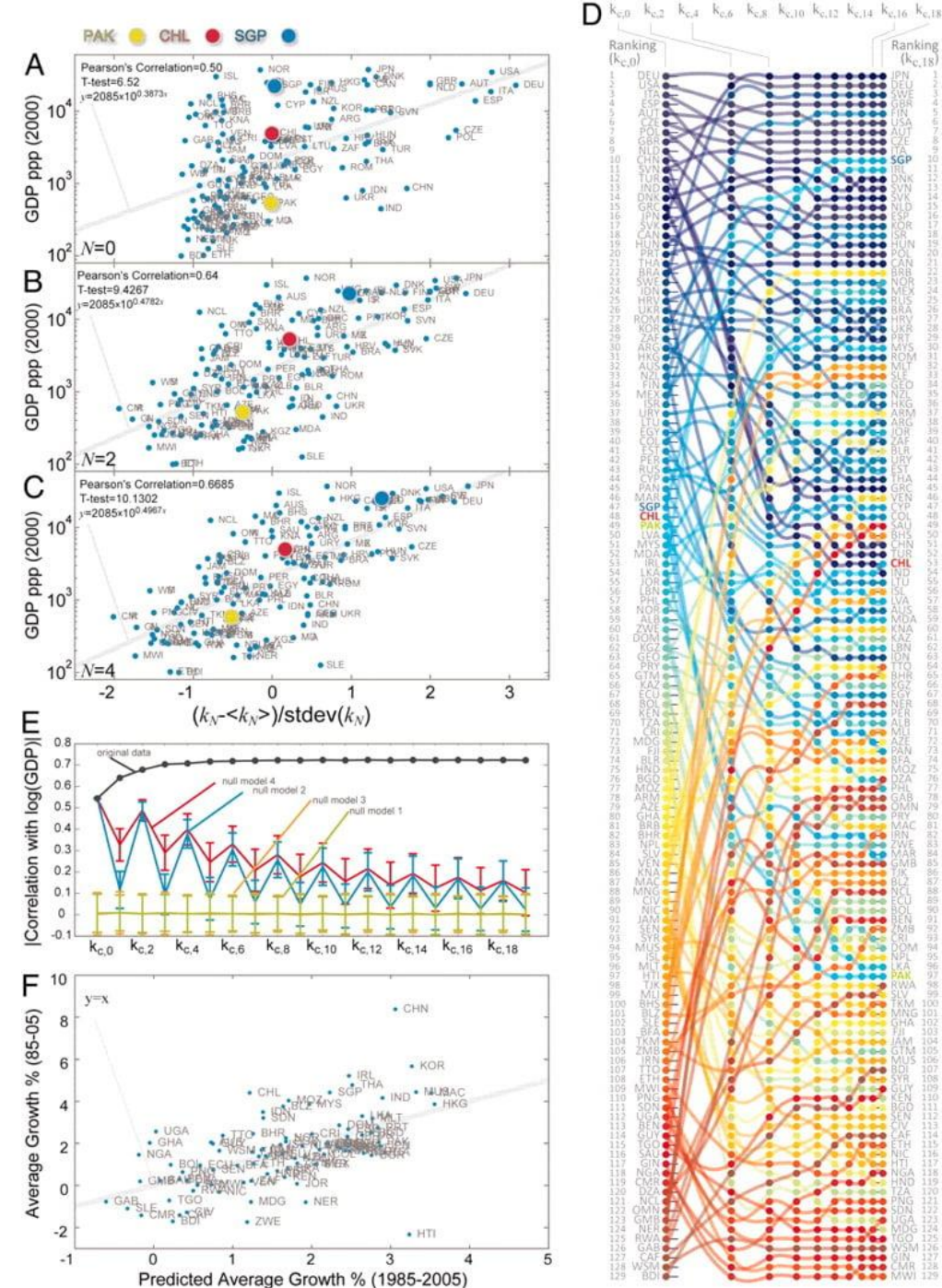
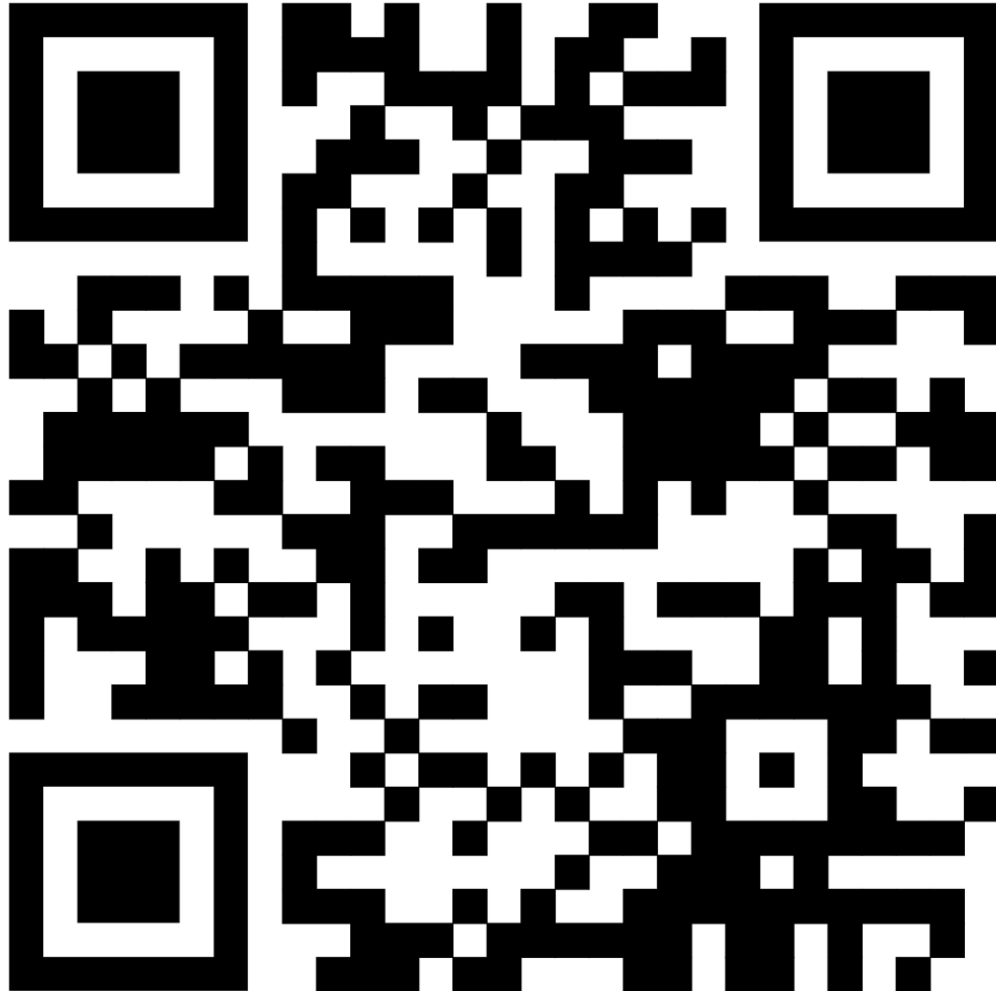
$$g_i = \beta_0 + \beta_1 \ln y_{0,i} + \beta_2 \ln(\text{ECI}_i) + \beta_3 \ln y_{0,i} \ln(\text{ECI}_i) + \varepsilon_i$$

Model estimates (Table 1) indicate that for trade and technology, the log-ECI main effect is positive and statistically significant at the 5% level of significance (Trade: $\hat{\beta}_2 \approx 4.63$, Technology: $\hat{\beta}_2 \approx 3.33$). In both models, the interaction term is negative and significant at the 10% level of significance (Trade: $\hat{\beta}_3 \approx -0.26$; Technology: $\hat{\beta}_3 \approx -0.20$), implying that the association between complexity and predicted growth attenuates as initial income rises. The convergence term is negative and highly significant in both cases ($\hat{\beta}_1 < 0$). By contrast, in the research model, neither the log-ECI main effect nor its interaction is statistically distinguishable from zero, and overall fit is markedly weaker (adjusted $R^2 \approx 0.35$). These patterns are consistent with the fitted lines: steep for Trade and Technology, comparatively flat for Research (though the slope does not matter here as it is statistically insignificant).

	Trade	Tech	Research
Intercept	18.86	16.85	12.38
$\ln(\text{ECI})$	4.62***	3.33***	4.04
$\ln(y_0)$	-1.56***	-1.40***	-1.00***
$\ln(y_0) * \ln(\text{ECI})$	-0.26*	-0.20*	-0.32
Adj. R^2	0.83	0.80	0.35
N	87	87	87

Table 1 * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Quiz 2 – September 30



Reading for Oct 7

Audretsch and Feldman 1996